GRADE 6

Computational Thinking

National Level Examination

SUBJECT CODE | | | |

Sample paper

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The Omr Answer Sheet.**
- **Do Not Forget To Sign The Omr Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And Hb Pencil To Fill The Omr Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Abstract & Spatial Thinking (10 Questions); Section B: Pattern Recognition (10 Questions); Section C: Decomposition, Data & Constraints (10 Questions); Section D: Algorithmic Thinking & Logical Flow (10 Questions); Section E: Ai Literacy & Olympiad Challenge (10 Questions).**
- **Only Omr Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The Omr Sheet And The Question Paper To The Invigilator.**

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. A large cube is painted on all six faces and then cut into 27 equal smaller cubes. How many small cubes have exactly two painted faces?
A. 10
B. 12
C. 16
D. 20

Q2. A square sheet is folded in half vertically, then in half horizontally, and then once along a diagonal of the folded square. One hole is punched at a point that is not on any fold. How many holes will appear when the sheet is fully opened?
A. 4
B. 6
C. 8
D. 16

Q3. A robot is facing North-West. It turns 135 degrees clockwise without moving. Which direction is it now facing?
A. South
B. East
C. North-East
D. South-East

Q4. How many lines of symmetry does a regular hexagon have?
A. 2
B. 3
C. 4
D. 6

Q5. An arrow points North-East. It is rotated 270 degrees counter-clockwise. In which direction will it point?
A. North-West
B. South-West
C. South-East
D. North

Q6. Five identical cubes are joined face-to-face in a straight line. How many faces of the cubes remain exposed altogether?
A. 18
B. 20
C. 22
D. 24

Q7. A cube is divided into 64 equal smaller cubes (4 x 4 x 4) after its entire outer surface is painted. How many small cubes have exactly one painted face?
A. 8
B. 16
C. 24
D. 32

Q8. On a map, 1.5 cm represents 6 km. Two towns are 7.5 cm apart on the map. What is their actual distance?
A. 24 km
B. 28 km
C. 30 km
D. 32 km

Q9. The side of a square is tripled. By what factor does its area increase?
A. 3
B. 6
C. 9
D. 12

Q10. A rectangular grid contains 4 rows and 6 columns of equal squares. How many grid intersection points are there, including the boundary points?
A. 28
B. 30
C. 35
D. 42

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 2, 6, 15, 31, 56, ___.
A. 81
B. 90
C. 92
D. 96

Q12. Find the next number: 3, 8, 18, 38, 78, ___.
A. 156
B. 158
C. 160
D. 162

Q13. Which letter comes next in the pattern? B, E, J, Q, ___.
A. W
B. X
C. Y
D. Z

Q14. Find the next pair: A1, C4, F9, J16, ___.
A. N25
B. O25
C. O20
D. P25

Q15. Find the next number: 4, 12, 10, 30, 28, 84, 82, ___.
A. 164
B. 168
C. 244
D. 246

Q16. Find the next number: 1, 4, 10, 22, 46, ___.
A. 90
B. 92
C. 94
D. 96

Q17. Find the next number: 2, 5, 11, 23, 47, ___.
A. 93
B. 94
C. 95
D. 96

Q18. Find the next number: 5, 7, 14, 16, 32, 34, ___.
A. 64
B. 66
C. 68
D. 70

Q19. The pattern ▲, ○, ■, ◆, ★, ● repeats continuously. Which symbol will be at the 47th position?
A. ■
B. ◆
C. ★
D. ●

Q20. A number-pair machine follows one rule: (2, 3) -> 8, (4, 5) -> 24, (3, 6) -> 21, (5, 7) -> 40. What will (6, 8) become?
A. 48
B. 50
C. 54
D. 56

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SECTION C – DECOMPOSITION, DATA & CONSTRAINTS

Q21. A two-digit number is greater than 60 but less than 90. It is divisible by both 4 and 9. What is the number?
A. 68
B. 72
C. 76
D. 84

Q22. Meera has three Rs.20 coins, four Rs.10 coins and five Rs.5 coins. She exchanges one Rs.20 coin for two Rs.10 coins. She then spends three Rs.10 coins and two Rs.5 coins. How much money and how many coins are left?
A. Rs.80 and 8 coins
B. Rs.85 and 8 coins
C. Rs.85 and 9 coins
D. Rs.90 and 8 coins

Q23. A bus starts with 42 passengers. At Stop 1, 8 get in and 5 get off. At Stop 2, 6 get in and 9 get off. At Stop 3, 10 get in and 4 get off. How many passengers are on the bus after Stop 3?
A. 44
B. 46
C. 48
D. 50

Q24. What is the smallest positive number that is exactly divisible by 6, 8 and 9?
A. 36
B. 48
C. 72
D. 144

Q25. A solid has two identical triangular faces and three rectangular faces. How many edges does the solid have?
A. 6
B. 8
C. 9
D. 12

Q26. A basket contains 48 fruits. Half are apples, one-third are oranges, and the rest are guavas. How many guavas are there?
A. 6
B. 8
C. 10
D. 12

Q27. A sensor records the values 12, 18, 15 and 21. An alert is generated for every reading greater than the average of all four readings. How many alerts are generated?
A. 1
B. 2
C. 3
D. 4

Q28. Among the integers from 1 to 30, how many are divisible by 2 or 3, but not by both?
A. 10
B. 12
C. 15
D. 20

Q29. In a group of 40 students, 22 learn Coding, 18 learn Robotics and 8 learn both. How many students learn neither Coding nor Robotics?
A. 6
B. 8
C. 10
D. 12

Q30. Five tasks S, P, Q, R and T must be completed. S is immediately before P. P is before Q. R is after Q. T is after R. Which task must be third?
A. P
B. Q
C. R
D. T

SECTION D – ALGORITHMIC THINKING & LOGICAL FLOW

Q31. A number follows this rule: if it is even, divide it by 2; if it is odd, multiply it by 3 and add 1. Starting with 14, apply the rule four times. What is the final number?
A. 17
B. 22
C. 34
D. 68

Q32. A machine takes an input, multiplies it by 2, adds 5, and then subtracts 3. If the output is 26, what was the input?
A. 10
B. 11
C. 12
D. 13

Q33. Start with 2. Repeat these two steps three times: (1) add 3; (2) multiply the result by 2. What is the final value?
A. 46
B. 50
C. 54
D. 58

Q34. A robot moves 4 squares North, 3 squares East, 2 squares South and 1 square West. In which direction is its final position from its starting point?
A. North-East
B. North-West
C. South-East
D. South-West

Q35. Start with a total of 0. For each number from 1 to 10, add the number if it is even and subtract it if it is odd. What is the final total?
A. -5
B. 0
C. 5
D. 10

Q36. An algorithm to find the perimeter of a rectangle has these steps: (1) read length and width; (2) sort the two values from smaller to larger; (3) compute 2 x (length + width); (4) display the result. Which step is unnecessary for finding the perimeter?
A. Step 1
B. Step 3
C. Step 2
D. Step 4

Q37. Variables A and B contain 4 and 7 respectively. The instructions are: C = A; A = B; B = C. What are the final values of A and B?
A. A = 4, B = 7
B. A = 7, B = 4
C. A = 11, B = 0
D. A = 4, B = 4

Q38. A program awards Gold for scores 80 or above, Silver for scores from 60 to 79, and Bronze for lower scores. What will a student scoring 75 receive?
A. Gold
B. Silver
C. Bronze
D. No award

Q39. A robot may jump either 2 squares or 5 squares in one move. What is the minimum number of jumps needed to travel exactly 19 squares?
A. 4
B. 5
C. 6
D. 7

Q40. Five tasks A, B, C, D and E obey these rules: A is before C; B is after A; D is after B; E is after both C and D. Which sequence satisfies all the rules?
A. A, B, D, C, E
B. A, C, E, B, D
C. B, A, C, D, E
D. A, D, B, C, E

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SECTION E – AI LITERACY & OLYMPIAD CHALLENGE

Q41. Which example is best described as fixed-rule automation rather than Artificial Intelligence?

A. A voice assistant interpreting spoken questions
B. A recommendation system learning a user's preferences
C. A street light programmed to switch on every day at 6:30 p.m.
D. A system identifying objects in photographs

Q42. Which activity is an example of supervised learning?

A. Grouping unlabeled songs by similarity
B. Learning from photographs already labelled 'cat' or 'dog'
C. A robot receiving rewards for reaching a target
D. Arranging random files by size

Q43. Which activity is an example of unsupervised learning?

A. Grouping unlabeled songs according to similar features
B. Training with labelled images of apples and oranges
C. Following a fixed timer
D. Learning by receiving reward points after each move

Q44. Which situation best represents reinforcement learning?

A. A spreadsheet calculating totals using a formula
B. A model trained on labelled examples
C. A system grouping similar items without labels
D. A game-playing agent receiving rewards for good moves and penalties for poor moves

Q45. A recorded voice note is primarily an example of which type of data?

A. Text data
B. Image data
C. Sound data
D. Numerical data

Q46. Which action best protects a student's online privacy?

A. Using the same easy password for every account
B. Using a strong unique password and keeping it private
C. Posting a password only to close friends
D. Writing the password publicly so it is not forgotten

Q47. Which statement about a digital footprint is most accurate?

A. It exists only while a device is switched on
B. It includes only files stored on a computer
C. Online actions can leave records that may remain even after content is deleted
D. It is created only when someone buys something online

Q48. A cycle-rental counter records 20, 25, 30 and 35 rentals on four consecutive days. If the same pattern continues, what is the most reasonable prediction for the fifth day?

A. 36
B. 38
C. 39
D. 40

Q49. A four-digit code uses different digits. The first digit is twice the last digit. The second digit is one more than the first. The third digit is three less than the second. The sum of all four digits is 20. What is the code?
A. 6543
B. 6743
C. 6752
D. 7643

Q50. A robot must travel from (0,0) to (3,2) using only one-step moves Right or Up. It must use a shortest path and it is not allowed to pass through (1,1). How many different shortest paths are possible?

A. 4
B. 6
C. 8
D. 10

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